Fatemeh Doudi

Ph.D. Student in Electrical Engineering
College Station, TX 77801
(+1) 979-422-1995 fatemehdoudi@tamu.edu

LinkedIn GitHub Google Scholar Personal Website

Education

Ph.D. in Computer Engineering
Texas A&M University Jan 2023 – Present
Advisor: Prof. Dileep Kalathil
GPA: 4.0/4.0

M.Sc. in Electrical Engineering
Sharif University of Technology Oct 2020 – Dec 2022
GPA: 17.78/20

B.Sc. in Electrical Engineering
Sharif University of Technology Sep 2015 – Dec 2019
GPA: 16.77/20

Research Experience

LLM-Driven Evolutionary Program Optimization
Texas A&M University May 2026 – Present

- Investigating agentic frameworks that leverage LLMs as evolutionary mutation operators for automated program optimization and open-ended scientific discovery.
- Studying the role of bandit algorithms in LLM ensemble selection within evolutionary frameworks, and analyzing sample-efficient search strategies including adaptive sampling and novelty-guided filtering.

Reinforcement Learning for Diffusion LLMs with Entropy-Guided Step Selection and Stepwise Advantages
Texas A&M University Aug. 2025 – Present
Under Review (Neurips)

- Formulated diffusion LLM generation as a finite-horizon MDP over denoising steps and derived an exact policy gradient with stepwise advantages.
- Proposed Entropy-Guided Stepwise Policy Optimization (EGSPO) for compute-efficient RL fine-tuning of diffusion LMs.
- Achieved consistent improvements on coding and reasoning benchmarks (MBPP, HumanEval, Sudoku, GSM8K, MATH500).

Inference-Time Multi-Preference Alignment for Diffusion Models
Texas A&M University Dec. 2024 – May 2025
Supervisors: Prof. Dileep Kalathil, Prof. P. R. Kumar
Accepted to ICLR 2026

- Proposed Diffusion Blend, an inference-time alignment framework enabling multi-objective reward blending without retraining.
- Achieved fine-tuned-level alignment performance while reducing alignment compute cost by 40
- Implemented and benchmarked gradient guidance, inference search, and KL-controlled reward blending methods.
- Demonstrated practical feasibility of post-training alignment for large diffusion models.

Time Series Prediction in Electric Power Systems using Deep State Space Model
Texas A&M University July. 2024 – Dec. 2024
Supervisors: Prof. Dileep Kalathil, Prof. Le Xie
Published in IEEE Transactions on Power Systems

- Designed PowerMamba, a deep state-space model for electric grid time-series forecasting.
- Achieved 7% accuracy improvement with 43% fewer parameters compared to transformer baselines.
- Implemented large-scale training and evaluation pipelines for spatio-temporal forecasting.

Exploring Large Language Models in the Electric Energy Sector
Texas A&M University Jan 2024 – Mar 2024
Supervisors: Prof. Dileep Kalathil, Prof. Le Xie
Published in Joule

- Investigated the integration of LLMs for data analysis and decision support in power grid operations.
- Gained experience in retrieval-augmented generation (RAG) and applying multimodal LLMs for fault detection and condition analysis in electric grids.

Skills

- **Programming Languages:** Python (primary), C/C++, MATLAB.
- **Machine Learning & LLMs:** Diffusion language models, autoregressive LLMs, reinforcement learning, RLHF, inference-time alignment, reasoning, preference optimization, retrieval-augmented generation (RAG).
- **Modeling & Architectures:** State-space models (Mamba), transformers, diffusion models, bandits, stochastic processes.
- **Systems & Tooling:** PyTorch, distributed training, large-scale experimentation, model evaluation and benchmarking, experiment tracking.
- **Theory:** Reinforcement learning, stochastic systems, game theory, convex and non-convex optimization.

Publications

Reinforcement Learning for Diffusion LLMs with Entropy-Guided Step Selection and Stepwise Advantages
V. T. Kunde*, **F. Doudi***, M. Farahbakhsh, D. Kalathil, K. Narayanan, J.-F. Chamberland
arXiv preprint, 2026. (*equal contribution)

Exploring the Capabilities and Limitations of Large Language Models in the Electric Energy Sector
S. Majumder*, L. Dong*, **F. Doudi***, Y. Cai*, C. Tian, D. Kalathil, K. Ding, A. A. Thatte
Joule, 2024. (*equal contribution)

PowerMamba: A Deep State Space Model and Comprehensive Benchmark for Time Series Prediction in Electric Power Systems
A. Menati, **F. Doudi**, D. Kalathil, L. Xie
IEEE Transactions on Power Systems, 2025

Diffusion Blend: Inference-Time Multi-Preference Alignment for Diffusion Models
M. Cheng, **F. Doudi**, D. Kalathil, M. Ghavamzadeh, P. R. Kumar
ICLR 2026 (accepted)
arXiv:2505.18547

Impact of Simulated Climate Data on Wind Power Prediction and Long-Term Grid Planning
F. Doudi, X. Liu, A. Menati, D. Fu, P. Chang, L. Xie
56th North American Power Symposium (NAPS), 2024

Applied Projects & Hackathons

AITX Techweek Hackathon – Agentic AI for Healthcare Matching
Austin,TX - GitHub Nov. 2025

- Built an end-to-end agentic AI pipeline for provider matching using NVIDIA Nemotron.
- Combined rule-based scoring with LLM reasoning to produce interpretable recommendations.
- Implemented a modular reasoning graph with asynchronous execution using LangGraph.

AITX DGX Spark Frontier Hackathon – Traffic Incidents & Insights (1st Place)
Austin, TX – GitHub Dec. 2025

- Developed a city-scale decision system for traffic incident prediction and response optimization.
- Trained a Mamba-based model on historical traffic data to generate spatio-temporal risk heatmaps.
- Trained and deployed models on NVIDIA DGX Spark hardware under real-time constraints.

Invited Talks & Presentations

- **Impact of Climate Data on Wind Power Prediction and Grid Planning**
  Oral Presentation, *56th North American Power Symposium (NAPS)*, 2024.

Professional Service

- Reviewer, *IEEE Power Engineering Letters*, 2025
- Reviewer, *Conference on Decision and Control (CDC)*, 2025
- Reviewer, *Transactions on Machine Learning Research (TMLR)*, 2025
- Reviewer, *ICML*, 2026
- Reviewer, *NeurIPS*, 2026

Relevent Courses

- **Machine Learning & AI:** ML, Deep Learning, Reinforcement Learning, Computer Vision
- **Optimization & Algorithms:** Convex Optimization, Advanced Optimization, Analysis of Algorithms
- **Stochastic Systems & Theory:** Stochastic Systems, Game Theory, Queueing Theory, Bandit Algorithms

Achievements and Awards

- **ECEN Merit Fellowship**, Texas A&M University
- Ranked **4th nationwide** in M.Sc. National Entrance Exam (20,000 participants)
- Ranked **57th nationwide** in B.Sc. National Entrance Exam (180,000 participants)